RV College of Engineering

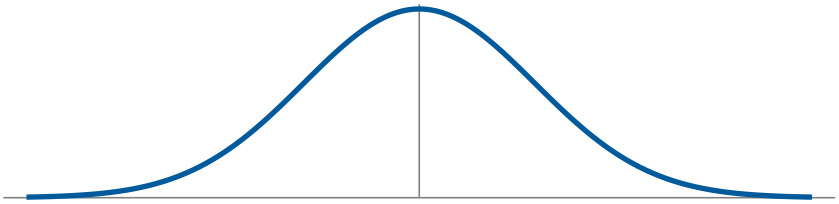
Department of Industrial Engineering and Management

Statistics for Data Analytics

Course Code: IM241AT

Semester IV • Professional Core Course

Unit 4



A Student Compilation for Study and Reference

Semester: IV						
STATISTICS FOR DATA ANALYTICS **Category: Professional Core Course** **(Theory)**						
Course Code	**:**	**IM241AT**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Data Summary and Presentation: Data types, tabular and graphical displays: Stem and Leaf diagrams, Histograms, Box plots, Radar diagrams. **Concepts of Probability:** Sample spaces and Events, Interpretations of probability, Addition rules, Conditional probability, Multiplication and Total probability rules, Independence, Bayes Theorem. Numerical Problems	
Unit-II	**09 Hrs**
Concepts of Random Variables: Random Variables, Discrete and continuous random variables. Probability distributions and mass functions, Numerical Problems **Discrete Probability Distributions:** Discrete uniform, Binominal, Poisson, Geometric, Negative binomial, Applications, Numerical Problems.	
Unit-III	**09 Hrs**
Continuous Probability Distributions: Continuous uniform, Normal, Normal approximations, Exponential, Applications, Numerical Problems. **Sampling Distribution:** Sampling distribution, Central Limit Theorem, Sampling distribution of means and derived quantities, Numerical Problems.	
Unit-IV	**09 Hrs**
Simple Linear Regression and Correlation: Empirical models, Simple Linear Regression, Properties of Least square Estimators and Estimation of variances, Common abuses of regression, Prediction of new observations, Correlation, Numerical Problems. Interpretation of graphical output from software packages such as Minitab	
Unit-V	**09 Hrs**
Statistical Inference for a single sample: Hypothesis testing, Inference on the mean of a normal population (variance known and unknown), Inference on the variance of a Normal population, Testing for Goodness of Fit, Tests of association, Numerical Problems **Interval Estimation:** Confidence Intervals on mean (variance known and unknown), and variance of a normal population	

Course Outcomes: After completing the course, the students will be able to	
CO1	Describe and report data set using data analysis, presentation and interpretation techniques to understand various phenomena in the fields of science and engineering.
CO2	Apply various statistical processing techniques to handle a set of data to estimate probabilities.
CO3	Apply an appropriate statistical tool and analyze a specific set of data to estimate and draw conclusions about population parameters
CO4	Draw inferences about population parameters and relations between variables based on analysis of sample data

References Books	
1.	Engineering Statistics, Douglas C. Montgomery, George C. Runger, Norma FarisHubele, 5th Edition, 2011, John Wiley & Sons, Inc., ISBN-13: 978- 0-470-63147-8
2.	Applied statistics and Probability for Engineers, Douglas C Montgomery, George C Runger, Wiley, 4th Edition, 2007, Asia Student Edition, ISBN: 978-81-265-2315-3.
3.	Statistics for Management, Richard I Levin, David S Rubin, 7th Edition, 1997, Prentice Hall India, ISBN: 9780134762920.
4.	Probability and Statistics for Engineers and Scientists, Walpole, Myers, Myers, Ye, 8th Edition, 2007, Pearson Education Inc., ISBN: 978-81-317-1552-9.
5.	Softwares : Microsoft Excel / Minitab / Matlab / R

6.	Online resources: a) http://172.16.44.44/nnptel.html - choose NOC:Introduction to Data Analytics(Course sponsored by Aricent) b) https://onlinecourses.nptel.ac.in/noc22_mg31/preview -Introduction to probability and Statistics c) https://newonlinecourses.science.psu.edu/statprogram/undergraduate-studies d) https://www.khanacademy.org/math/statistics-probability

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40**	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: (Internal Choice)	16
5 & 6	Unit 3: (Internal Choice)	16
7 & 8	Unit 4: (Internal Choice)	16
9 & 10	Unit 5: (Internal Choice)	16
	TOTAL	**100**

A Note on This Compilation

This book is **not** the official RV College of Engineering notes. It is a compilation of student notes together with the handwritten notes provided by **Dr. Ramaa A Nadig**, Department of Industrial Engineering and Management.

All information contained in this book has been drawn from her notes and lectures. The material has been organised, typeset, and arranged to follow the prescribed course syllabus, but it should be treated as a supplementary study aid rather than an official course document.

Every effort has been made to reproduce the source notes faithfully and to preserve their content without addition or deletion. Where the original handwritten working was unclear, the final answers recorded in the notes were treated as authoritative. Readers should always cross-check against the prescribed textbooks and their instructor's guidance.

Compiled from the notes and lectures of Dr. Ramaa A Nadig, Department of Industrial Engineering and Management.

Contents

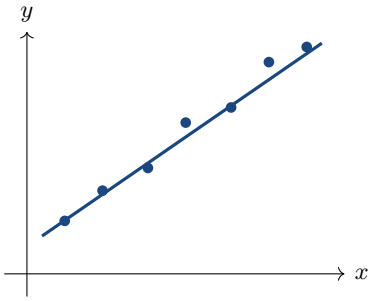
Front Matter

Cover Page 1
Syllabus (Official Course Sheet) 2
A Note on This Compilation 4
Contents 5

Unit 4

Chapter 1. The Simple Linear Regression Model 7
Chapter 2. Least Squares Estimation 10
Chapter 3. Estimating the Error Variance 13
Chapter 4. Properties, Inference and Prediction 14
Chapter 5. Regression Variants and Correlation 17
Chapter 6. Worked Problems 21
Chapter 7. Formula Summary 28

Unit 4



Chapter 1

The Simple Linear Regression Model

1.1 Regression Analysis

Regression Analysis

Regression analysis is a statistical technique that is useful for exploring the relationships between two or more variables.

In the simple linear regression model the *dependent variable*, or *response*, is related to one *independent* or *regressor* variable as

Simple Linear Regression Model

$$Y = \beta_0 + \beta_1 x + \epsilon,$$

where ϵ is the random error term. The parameters β_0 and β_1 are called the *regression coefficients*.

In a scatter plot,

- $x \rightarrow$ the independent or regressor variable;
- $Y \rightarrow$ the response or dependent variable.

We estimate β_0 and β_1 by maximum likelihood, by the properties of least squares, by the method of moments, and so on. The method of maximum likelihood needs further knowledge of the distribution of Y, whereas the method of moments and the principle of least squares do not need any knowledge about the distribution of Y.

1.2 The Meaning of "Regression"

The word *regression* means "to move in the backward direction." A model exists in nature but is unknown to the experimenter. When some values of the explanatory variables are provided, then the values for the output or study variable are generated depending on the values of the explanatory variables. So the form of the function and the nature of the phenomenon give rise to the data. Ideally, the pre-existing model gives rise to the data.

Our objective is to determine the final form of this model. We propose to first move in the backward direction: we collect the data on the study and explanatory variables, then employ some statistical techniques and use this data to know the form of the function f. Equivalently, the data from the model is recorded first and then used to determine the parameters of the

model. The regression analysis is the technique which helps in determining the statistical model by using the data on study and explanatory variables.

1.3 Steps in Regression Analysis

The stages of a regression study proceed as follows:

1. Statement of the problem under consideration.
2. Choice of relevant variables.
3. Collection of data on the relevant variables.
4. Specification of the model.
5. Choice of method for fitting the data.
6. Fitting of the model.
7. Model validation and criticism.
8. Using the chosen model(s) for the solution of the posed problem and for forecasting.

1.4 The Mean and Variance of the Response

Based on the problem and the scatter diagram, it is reasonable to assume that the mean of the random variable Y is related to x by the following straight-line relation:

True Regression Line (line of mean values)

$$\mathbb{E}(Y \mid x) \;=\; \mu_{Y|x} \;=\; \beta_0 + \beta_1 x,$$

where the slope β_1 and intercept β_0 of the line are called the regression coefficients.

We think of the regression model as an *empirical* model. Suppose that the mean and variance of ϵ are 0 and σ^2 respectively. Then

$$\mathbb{E}(Y \mid x) \;=\; \mathbb{E}\big(\beta_0 + \beta_1 x + \epsilon\big) \;=\; \beta_0 + \beta_1 x + \mathbb{E}(\epsilon) \;=\; \beta_0 + \beta_1 x.$$

The variance of Y given x is

$$\mathrm{Var}(Y \mid x) \;=\; \mathrm{Var}\big(\beta_0 + \beta_1 x + \epsilon\big) \;=\; \mathrm{Var}(\beta_0 + \beta_1 x) + \mathrm{Var}(\epsilon) \;=\; 0 + \sigma^2 \;=\; \sigma^2.$$

The true regression model is a line of mean values:

$$\mu_{Y|x} = \beta_0 + \beta_1 x.$$

Here β_1 can be interpreted as the change in the mean of Y for a unit change in x. Also, the variability of Y at a particular value of x is determined by the error variance σ^2. This implies that there is a distribution of Y values at each x, and that the variance of this distribution is the same at each x.

Simple Linear Regression Model (full statement)

The model

$$Y = \beta_0 + \beta_1 x + \epsilon$$

is called the *simple linear regression model*. The true regression $\mu_{Y|x} = \beta_0 + \beta_1 x$ is a line of mean values: the height of the regression line at any value of x is just the expected value of Y for that x. The slope β_1 is the change in the mean of Y for a unit change in x. The variability of Y at a particular value of x is determined by the error variance σ^2; there is a distribution of Y values at each x and the variance of this distribution is the same at each x.

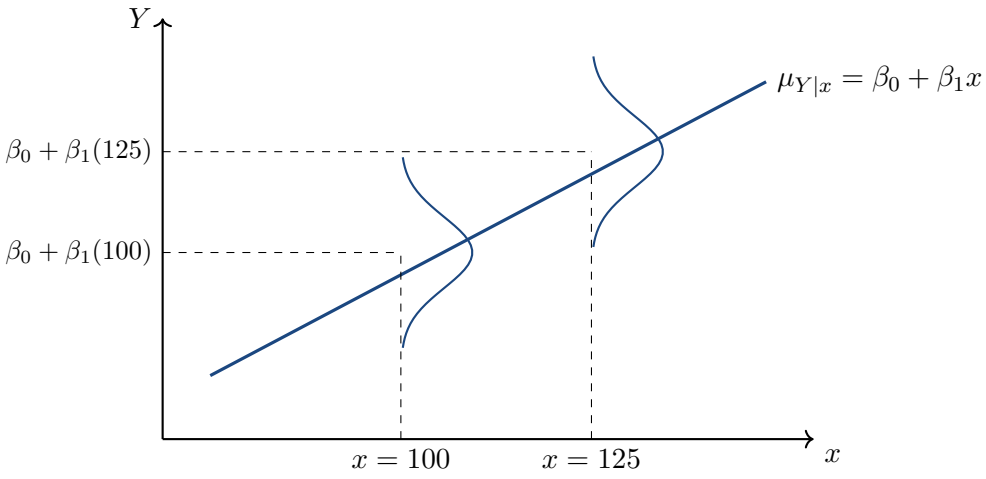


Figure 1.1: The true regression line is a line of mean values. At each x there is a distribution of Y with the same variance σ^2.

Chapter 2

Least Squares Estimation

Consider the model

$$\mathbb{E}(Y \mid x) = \beta_0 + \beta_1 x,$$

where the intercept β_0 and the slope β_1 are unknown regression coefficients. We assume that each observation Y_i can be described by the model

$$Y = \beta_0 + \beta_1 x + \epsilon,$$

where ϵ is a random error with mean zero and (unknown) variance σ^2.

2.1 The Method of Least Squares

The German scientist Karl Gauss (1777–1855) proposed estimating the parameters β_0 and β_1 so as to minimize the sum of squares of the vertical deviations. This is called the *method of least squares*.

Using the model for the i-th observation,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad i = 1, 2, \ldots, n, \tag{2.1}$$

the sum of squares of the deviations of the observations from the true regression line is

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} \big(y_i - \beta_0 - \beta_1 x_i\big)^2. \tag{2.2}$$

The least squares estimators of β_0 and β_1, say $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

$$\left.\frac{\partial L}{\partial \beta_0}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^{n} \big(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\big) = 0, \tag{2.3}$$

$$\left.\frac{\partial L}{\partial \beta_1}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^{n} \big(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\big) x_i = 0. \tag{2.4}$$

Simplifying these two equations yields the *least squares normal equations*:

Least Squares Normal Equations

$$\left.\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i &= \sum_{i=1}^{n} y_i, \\ \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} x_i y_i. \end{aligned}\right\} \tag{2.5}$$

2.2 The Least Squares Estimators

Solving the normal equations gives the least squares estimates of the intercept and slope:

Least Squares Estimators

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \tag{2.6}$$

$$\hat{\beta}_1 = \frac{\displaystyle\sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\displaystyle\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}, \tag{2.7}$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

The fitted or estimated regression line is

Fitted Regression Line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Note that each pair of observations satisfies the relationship

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i, \qquad i = 1, 2, \ldots, n,$$

where $e_i = y_i - \hat{y}_i$ is called the *residual*. The residual describes the error in the fit of the model to the i-th observation y_i.

Residual

The residual for observation i is

$$e_i = y_i - \hat{y}_i.$$

It measures the error in the fit of the model to the i-th observed value.

2.3 Corrected Sums of Squares

Notationally, it is occasionally convenient to give special symbols to the numerator and denominator of Equation (2.7):

S_{xx} and S_{xy}

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}, \tag{2.8}$$

$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})\, y_i = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}. \tag{2.9}$$

With these, $\hat{\beta}_1 = \dfrac{S_{xy}}{S_{xx}}$.

Chapter 3

Estimating the Error Variance

There is actually another unknown parameter in the regression model, namely σ^2 (the variance of the error term ϵ). The residuals $e_i = y_i - \hat{y}_i$ are used to estimate σ^2.

3.1 The Error Sum of Squares

The sum of squares of the residuals, often called the *error sum of squares*, is

Error Sum of Squares

$$\mathrm{SS_E} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2. \tag{3.1}$$

It can be shown that the expected value of the error sum of squares is

$$\mathbb{E}(\mathrm{SS_E}) = (n-2)\,\sigma^2.$$

Therefore an *unbiased estimator* of σ^2 is

Unbiased Estimator of σ^2

$$\hat{\sigma}^2 = \frac{\mathrm{SS_E}}{n-2}. \tag{3.2}$$

A convenient computational form of the error sum of squares is

Computing $\mathrm{SS_E}$

$$\mathrm{SS_E} = \mathrm{SS_T} - \hat{\beta}_1 S_{xy},$$

where

$$\mathrm{SS_T} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2$$

is the total (corrected) sum of squares of the response variable y.

Chapter 4

Properties, Inference and Prediction

4.1 Properties of the Least Squares Estimators

In simple linear regression the estimated standard error of the slope and the estimated standard error of the intercept are

Standard Errors of the Estimators

$$\text{se}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \qquad \text{se}(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)},$$

where $\hat{\sigma}^2 = \dfrac{\text{SS}_\text{E}}{n-2}$.

The covariance of the two estimators is

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2\,\bar{x}}{S_{xx}}.$$

4.2 Analysis of Variance Approach to Test Significance of Regression

A method called the *analysis of variance* can be used to test for significance of regression. The identity underlying the method is

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

The two components on the right-hand side measure, respectively, the amount of variability in y_i accounted for by the regression line, and the residual variation left unexplained by the regression line. Symbolically it can be rewritten as

Partition of Total Variability

$$\text{SS}_\text{T} = \text{SS}_\text{R} + \text{SS}_\text{E},$$

where

$$\text{SS}_\text{T} = \sum_{i=1}^{n}(y_i - \bar{y})^2, \quad \text{SS}_\text{R} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2, \quad \text{SS}_\text{E} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

It is seen that $\mathrm{SS_E} = \mathrm{SS_T} - \hat{\beta}_1 S_{xy}$, so that $\mathrm{SS_T} = \hat{\beta}_1 S_{xy} + \mathrm{SS_E}$. Hence the regression sum of squares is

$$\mathrm{SS_R} = \hat{\beta}_1 S_{xy}.$$

The total sum of squares $\mathrm{SS_T}$ has $n-1$ degrees of freedom, and $\mathrm{SS_R}$ and $\mathrm{SS_E}$ have 1 and $n-2$ degrees of freedom respectively. We may show that

$$\mathbb{E}\left[\frac{\mathrm{SS_E}}{n-2}\right] = \sigma^2, \qquad \mathbb{E}(\mathrm{SS_R}) = \sigma^2 + \beta_1^2 S_{xx},$$

and that $\mathrm{SS_E}/\sigma^2$ and $\mathrm{SS_R}/\sigma^2$ are independent chi-square random variables with $n-2$ and 1 degrees of freedom respectively.

Thus, if the null hypothesis $H_0 : \beta_1 = 0$ is true, the statistic

Test Statistic for Significance of Regression

$$F_0 = \frac{\mathrm{SS_R}/1}{\mathrm{SS_E}/(n-2)} = \frac{\mathrm{MS_R}}{\mathrm{MS_E}}$$

follows the $F_{1,n-2}$ distribution, and we would reject H_0 if $f_0 > f_{\alpha,1,n-2}$.

Here $\mathrm{MS_R}$ and $\mathrm{MS_E}$ are called *mean squares*. The test procedure is usually arranged in an analysis of variance table.

Table 4.1: Analysis of variance table for significance of regression.

Source of variation	Sum of squares	DoF	MS	F_0
Regression	$\mathrm{SS_R} = \hat{\beta}_1 S_{xy}$	1	$\mathrm{MS_R}$	$\mathrm{MS_R} / \mathrm{MS_E}$
Error	$\mathrm{SS_E} = \mathrm{SS_T} - \hat{\beta}_1 S_{xy}$	$n-2$	$\mathrm{MS_E}$	
Total	$\mathrm{SS_T}$	$n-1$		

where $\mathrm{MS_E} = \hat{\sigma}^2$.

4.3 Prediction of a New Observation

A $100(1-\alpha)\%$ *prediction interval* on a future observation at the value x_0 is given by

Prediction Interval for a Future Observation

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]} \;\le\; Y_0 \;\le\; \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]},$$

where the value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

4.4 Coefficient of Determination R^2

Coefficient of Determination

The quantity

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_E}}{\mathrm{SS_T}}$$

is called the *coefficient of determination*. It is often used to judge the adequacy of the regression model. R^2 is interpreted as the proportion of variability in the data explained (or accounted for) by the model.

4.5 Misinterpretation or Misuse of Linear Regression Techniques

It is possible to develop statistically significant relationships between variables that are completely unrelated in a causal sense. Common pitfalls include:

1. Prediction beyond the range of the observed x.
2. Prediction beyond the time period of data collection.
3. Equating a statistical relation to a real-life (causal) situation — confusing correlation with causation.
4. Misinterpreting R^2 to be the strength of the relationship instead of the proportion of variability explained.
5. Regression models are not necessarily valid for extrapolation purposes.

Chapter 5

Regression Variants and Correlation

5.1 Summary of the Model and Assumptions

Recall the simple linear regression model

$$Y = \beta_0 + \beta_1 x + \epsilon,$$

in which the slope β_1 and intercept β_0 are called the regression coefficients, and:

- $Y \to$ dependent or study variable;
- $x \to$ independent or explanatory variable;
- $\beta_0 \to$ intercept term;
- $\beta_1 \to$ slope parameter;
- $\epsilon \to$ error component, which accounts for the failure of the data to lie on the straight line and represents the difference between the true and observed values of y.

ϵ is assumed to be an i.i.d. random variable with mean zero and constant variance σ^2, and it is also assumed to be normally distributed. Consequently,

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_1 x, \qquad \mathrm{Var}(y \mid x) = \sigma^2.$$

5.2 Least Squares Estimation — Geometric View

The principle of least squares estimates the parameters β_0 and β_1 by minimizing the sum of squares of the differences between the observations and the line in the scatter diagram.

5.3 Direct, Reverse and Orthogonal Regression

Direct regression

When the vertical difference between observations and the line in the scatter diagram is considered and its sum of squares is minimized to obtain estimates of β_0 and β_1, the method is *direct regression*. This method is also known as *ordinary least squares estimation*. The direct regression approach minimizes the sum of squares

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2.$$

Taking partial derivatives and setting them to zero we obtain

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

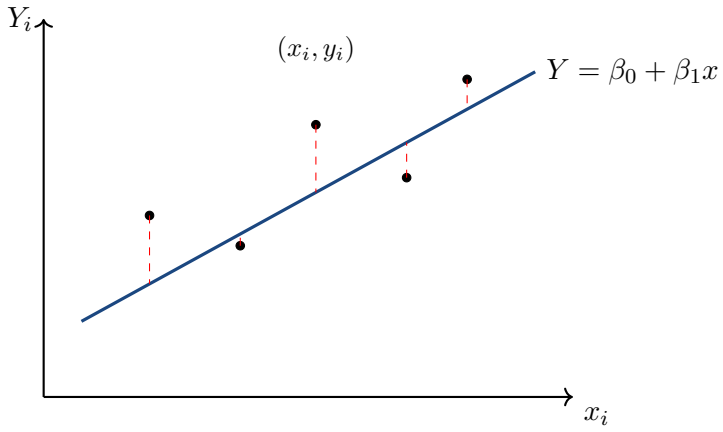


Figure 5.1: When the vertical differences between the observations and the line in the scatter diagram are considered and their sum of squares is minimized, the method is *direct regression*.

Reverse (inverse) regression

Alternatively, the sum of squares of the differences between observations and the line in the *horizontal* direction in the scatter diagram can be minimized to obtain estimates. This is known as the *reverse* (or inverse) regression method.

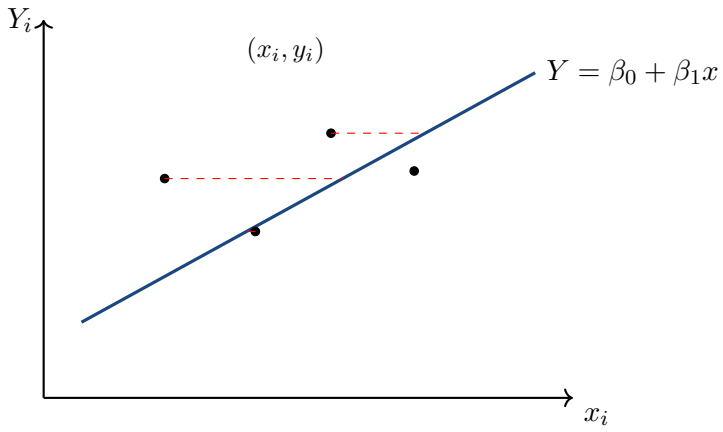


Figure 5.2: Reverse regression: horizontal deviations are minimized.

Orthogonal (major axis) regression

Instead of the horizontal or vertical direction, if the sum of squares of the *perpendicular* distances between the observations and the line in the scatter diagram is minimized, the method is known as *orthogonal regression* or the *major axis regression* method.

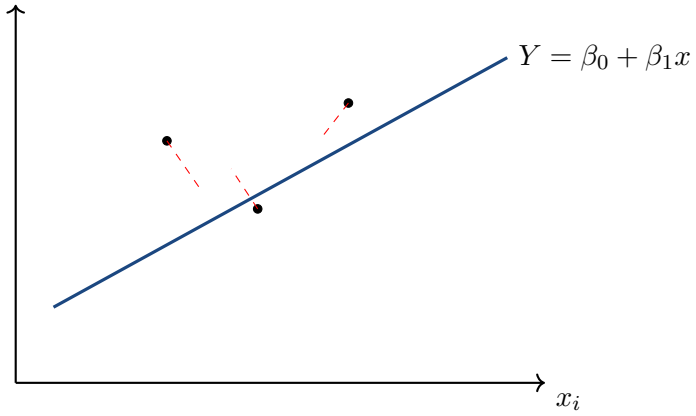


Figure 5.3: Major axis (orthogonal) regression: perpendicular distances are minimized.

5.4 Coefficient of Determination (restated)

Key Sums and Estimators

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}},$$

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - \frac{(\sum x_i)\,(\sum y_i)}{n},$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{(\sum x_i)^2}{n},$$

$$S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{(\sum y_i)^2}{n},$$

$$\hat{\sigma}^2 = \frac{\mathrm{SS_E}}{n-2}, \qquad \mathrm{SS_E} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \mathrm{SS_T} - \hat{\beta}_1 S_{xy}.$$

Here S_{xx} is (proportional to) the variability of x. The coefficient of determination

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_E}}{\mathrm{SS_T}}$$

is used to judge the adequacy of the model: it is the amount of variability in the data explained or accounted for by the model.

5.5 Correlation Coefficient

Population Correlation Coefficient

The correlation coefficient between Y and X is

$$\rho = \frac{\sigma_{xy}}{\sigma_x \, \sigma_y},$$

where σ_{xy} is the covariance between Y and X, σ_x is the standard deviation (spread) of X, and σ_y is the standard deviation of Y.

The *sample correlation coefficient* is

Sample Correlation Coefficient

$$R = \frac{\sum_{i=1}^{n} y_i (x_i - \bar{x})}{\left[\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2 \right]^{1/2}} = \frac{S_{xy}}{(S_{xx} \, \mathrm{SS_T})^{1/2}}.$$

Remark 5.1. The slope and the sample correlation coefficient are related by

$$\hat{\beta}_1 = \left(\frac{\mathrm{SS_T}}{S_{xx}} \right)^{1/2} R.$$

The sample correlation coefficient R measures the linear association between Y and X, while $\hat{\beta}_1$ measures the predicted change in the mean of Y for a unit change in X.

Chapter 6

Worked Problems

This chapter collects the worked numerical problems from the notes. Each uses the least squares machinery developed earlier: $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, and $\hat{\sigma}^2 = \mathrm{SS_E}/(n-2)$.

6.1 Roadway Surface Temperature and Pavement Deflection

Example 6.1 (Pavement deflection)**.** Regression methods were used to analyse the data from a study investigating the relationship between roadway surface temperature x and pavement deflection y. Summary quantities were

$$n = 20, \quad \sum y_i = 12.75, \quad \sum y_i^2 = 8.86, \quad \sum x_i = 1478,$$

$$\sum x_i^2 = 143{,}215.8, \quad \sum x_i y_i = 1083.67.$$

(a) Calculate the least squares estimates of the slope and intercept. Graph the regression line.

(b) Use the equation of the fitted line to predict the pavement deflection when the surface temperature is 85°F.

(c) What is the mean pavement deflection when the surface temperature is 90°F?

(d) What change in mean pavement deflection would be expected for a 1°F change in surface temperature?

Solution

(a) Using $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$,

$$S_{xx} = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n} = 143{,}215.8 - \frac{(1478)^2}{20} = 33{,}991.6,$$

$$S_{xy} = \sum x_i y_i - \frac{\left(\sum x_i\right)\left(\sum y_i\right)}{n} = 1083.67 - \frac{12.75 \times 1478}{20} = 141.445.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{141.445}{33{,}991.6} = 0.00416,$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{12.75}{20} - (0.00416)\frac{1478}{20} = 0.32999.$$

The fitted regression line is

$$\boxed{\hat{y} = 0.32999 + 0.00416\, x.}$$

We also estimate the error variance:

$$\hat{\sigma}^2 = \mathrm{MS_E} = \frac{\mathrm{SS_E}}{n-2} = \frac{\left(\sum y_i^2 - n\bar{y}^2\right) - \hat{\beta}_1 S_{xy}}{n-2}.$$

Now $\sum y_i^2 - n\bar{y}^2 = 8.86 - 20\left(\frac{12.75}{20}\right)^2$ and $\hat{\beta}_1 S_{xy} = 0.00416 \times 141.445$, so

$$\hat{\sigma}^2 = \frac{8.86 - 20\left(\frac{12.75}{20}\right)^2 - 0.00416 \times 141.445}{18} = \frac{0.143}{18} = 7.91 \times 10^{-3} \approx 0.00796.$$

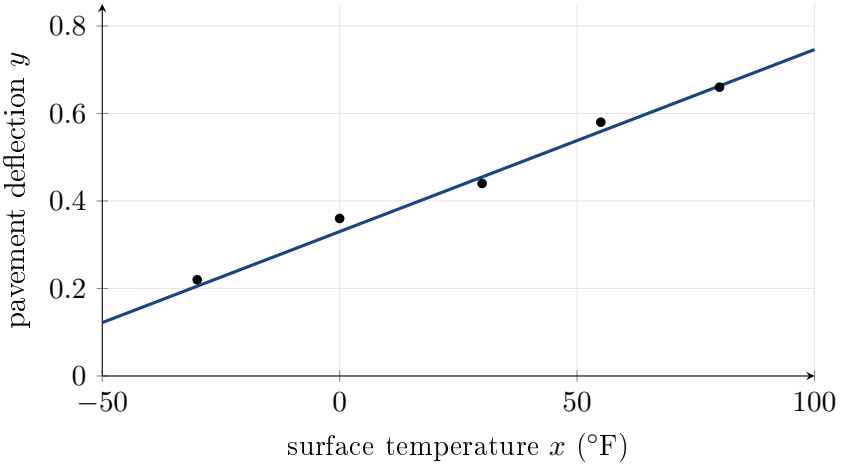


Figure 6.1: Fitted regression line $\hat{y} = 0.32999 + 0.00416x$ for the pavement deflection data (illustrative points shown).

(b) At $x = 85$°F,

$$\hat{y} = 0.32999 + 0.00416 \times 85 = 0.6836.$$

(c) At $x = 90$°F,

$$\hat{y} = 0.32999 + 0.00416 \times 90 = 0.7044.$$

(d) A change of 0.00416 in mean pavement deflection would be expected for a 1°F change in surface temperature, since 0.00416 is the slope, and the slope is the average increase/decrease of y per unit change of x.

6.2 Changing Units: Celsius Instead of Fahrenheit

Example 6.2 (Rescaling the regressor)**.** Suppose the temperature in the previous problem is measured in °C rather than °F.

(a) Write the new regression model that results.

(b) What change in expected pavement deflection is associated with a 1°C change in surface temperature?

Solution

(a) Using the conversion $x_F = \frac{9}{5}x_C + 32$ and substituting into the fitted model $\hat{y} = 0.3299982 + 0.0041612\,x_F$,

$$\hat{y} = 0.3299982 + 0.0041612\left(\tfrac{9}{5}x_C + 32\right),$$

$$\hat{y} = 0.3299982 + 0.0074902\,x_C + 0.1331584,$$

$$\boxed{\hat{y} = 0.4631476 + 0.0074902\,x_C.}$$

(b) The new slope is

$$\hat{\beta}_1 = 0.0074902,$$

so a 1°C change in surface temperature is associated with a change of 0.0074902 in expected pavement deflection.

6.3 Compressive Strength and Intrinsic Permeability of Concrete

Example 6.3 (Concrete permeability)**.** An article presented data on compressive strength x and intrinsic permeability y of various concrete mixes. Summary quantities are

$$n = 14, \quad \sum y_i = 572, \quad \sum y_i^2 = 23{,}530, \quad \sum x_i = 43,$$

$$\sum x_i^2 = 157.42, \quad \sum x_i y_i = 1697.80.$$

Assume the two variables are related according to the simple linear regression model.

(a) Calculate the least squares estimates of the slope and intercept.

(b) Use the fitted line to predict the permeability when the compressive strength is $x = 4.3$.

(c) Give a point estimate of the mean permeability when the compressive strength is $x = 3.7$.

(d) Suppose the observed value of permeability is $y = 46.1$ at $x = 3.7$. Calculate the corresponding residual.

Solution

Using $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$,

$$S_{xx} = 157.42 - \frac{43^2}{14} = 25.348571,$$

$$S_{xy} = 1697.80 - \frac{(43)(572)}{14} = -59.057143.$$

(a)

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-59.057143}{25.348571} = -2.330,$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{572}{14} - (-2.3298017)\frac{43}{14} = 48.013.$$

The fitted line is

$$\hat{y} = 48.013 - 2.3298017\,x.$$

(b) At $x = 4.3$,

$$\hat{y} = 48.013 - 2.3298017 \times 4.3 = 37.99.$$

(c) At $x = 3.7$,

$$\hat{y} = 48.013 - 2.3298017 \times 3.7 = 39.39.$$

(d) The residual at $x = 3.7$ with observed $y = 46.1$ is

$$e = y - \hat{y} = 46.1 - 39.39 = 6.71.$$

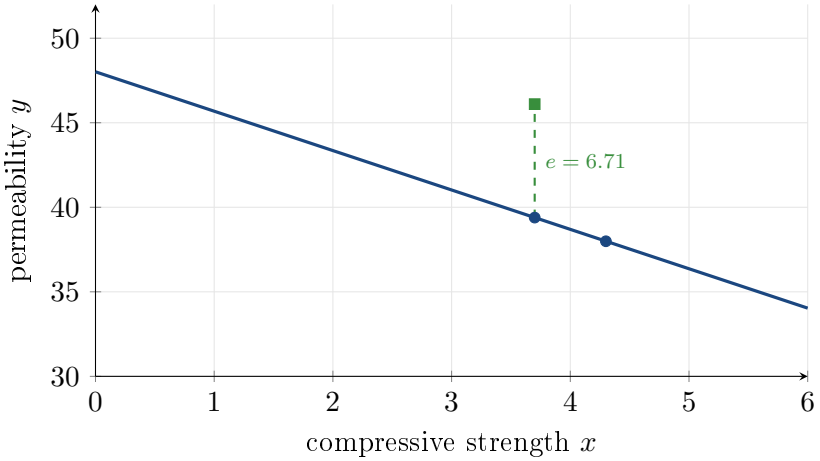


Figure 6.2: Fitted line $\hat{y} = 48.013 - 2.3298x$; the residual at $x = 3.7$ is the dashed segment.

6.4 Green Liquor Na_2S Concentration and Paper Machine Production

Example 6.4 (Green liquor)**.** Data on green liquor Na_2S concentration (y, in g/L) and paper machine production (x, in tons/day) are as follows:

y	40	42	49	46	44	48	46	43	53	52	54	57	58
x	825	830	890	895	890	910	915	960	990	1010	1012	1030	1050

(a) Fit a simple linear regression model with y = green liquor Na_2S concentration and x = production. Find an estimate of σ^2. Draw a scatter diagram of the data and the resulting least squares fitted model.

(b) Find the fitted value of y corresponding to $x = 910$ and the associated residual.

(c) Find the mean green liquor Na_2S concentration when the production rate is 950 tons/day.

Solution

With $n = 13$ the summary quantities are

$$\sum x_i^2 = 11{,}529{,}419, \quad \sum y_i^2 = 31{,}128, \quad \sum x_i = 12{,}207, \quad \sum y_i = 632, \quad \sum x_i y_i = 598{,}098.$$

(a)

$$S_{xx} = 11{,}529{,}419 - \frac{(12{,}207)^2}{13} = 67{,}046,$$

$$S_{xy} = 598{,}098 - \frac{(12{,}207)(632)}{13} = 4650.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{4650}{67{,}046} = 0.06936,$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 48.615 - 939 \times 0.06936 = -16.51.$$

The fitted model is

$$\boxed{\hat{y} = -16.51 + 0.06936\, x.}$$

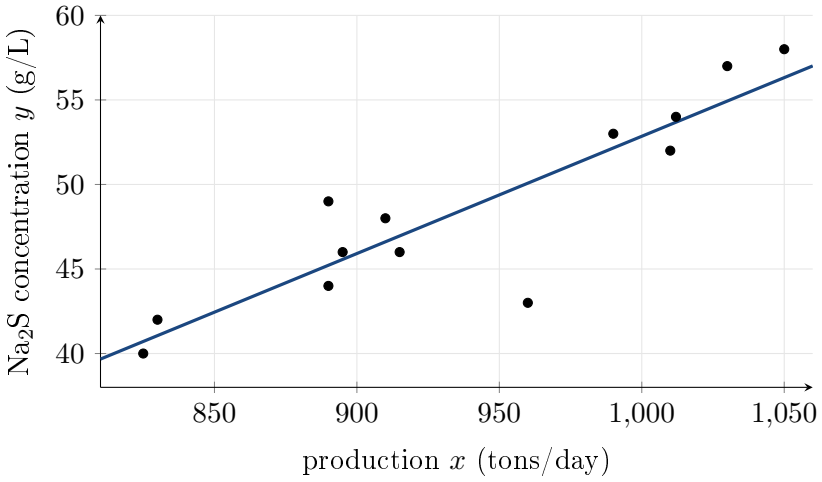


Figure 6.3: Scatter diagram of green liquor data with fitted line $\hat{y} = -16.51 + 0.06936x$.

(b) At $x = 910$,

$$\hat{y} = -16.51 + 0.06936 \times 910 = 46.60.$$

The observed value at this production rate is $y = 48$, so the residual is

$$e = y - \hat{y} = 48 - 46.60 = 1.3924.$$

(c) At $x = 950$,

$$\hat{y} = -16.51 + 0.06936 \times 950 = 49.38.$$

6.5 Correlation by Inspection and a Re-coded Observation

Example 6.5 (Inspection and recoding)**.** For a set of 20 pairs of values (x_i, y_i),

$$\sum x_i^2 = 130{,}000, \quad \sum y_i^2 = 88.6, \quad \sum x_i y_i = 1500,$$

$$\sum x_i = 1500, \quad \sum y_i = 10.$$

(a) From the above values, without deriving any other quantities, is it possible to tell whether x and y are truly correlated just by inspection? Give reasons. Also write the estimated intercept and slope parameters and the fitted linear regression model.

(b) If an observed (x, y) value which was wrongly recorded previously as $(25, 3)$ is changed to the correct observation $(35, 2)$, derive the changed regression model.

Solution

(a) The sign of S_{xy} determines positive or negative correlation. Now

$$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = \sum x_i y_i - \sum x_i\, \bar{y}.$$

Here $\sum x_i y_i = \sum x_i$ (both equal 1500) and $\bar{y} = \frac{10}{20} < 1$, so S_{xy} is positive. Therefore the correlation between x and y is positive.

Computing the estimates,

$$S_{xy} = 1500 - 1500 \times \tfrac{1}{2} = 750,$$

$$S_{xx} = 130{,}000 - \frac{(1500)^2}{20} = 17{,}500,$$

so the slope is

$$\hat{\beta}_1 = \frac{750}{17{,}500} = 0.0429,$$

and the intercept is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.5 - 0.0429 \times 75 = -2.7143.$$

(b) Replacing the observation $(25, 3)$ by $(35, 2)$, the modified summary quantities are

$$\sum x_i^2 = 130{,}000 - 25^2 + 35^2 = 130{,}600,$$

$$\sum y_i^2 = 88.6 - 3^2 + 2^2 = 83.6,$$

$$\sum x_i y_i = 1500 - 75 + 70 = 1495,$$

$$\sum x_i = 1500 - 25 + 35 = 1510, \qquad \bar{x} = 75.5,$$

$$\sum y_i = 10 - 3 + 2 = 9, \qquad \bar{y} = 0.45.$$

Then

$$S_{xy} = \sum x_i y_i - \frac{\sum x_i \sum y_i}{n} = 1495 - 20 \times 75.5 \times 0.45 = 815,$$

$$S_{xx} = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n} = 130{,}600 - \frac{(1510)^2}{20} = 16{,}595.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = 0.0491, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = -3.26,$$

and the changed regression model is

$$\boxed{\hat{y} = -3.26 + 0.0491\, x.}$$

6.6 Residual Analysis and Goodness of Fit

Example 6.6 (Residuals and R^2)**.** The following data were obtained in an experiment.

x_i	11	15	17	20	13	21
y_i	43.5	40	38.9	38.1	44	39

(a) Compute the residuals for all corresponding x_i, assuming a linear regression model.

(b) Draw a residual plot and interpret it.

(c) How close does the regression model fit the data? Answer with the help of a computed measure.

Solution

The summary quantities are

$$\bar{x} = 16.17, \quad \bar{y} = 40.58, \quad \sum x_i^2 = 1645, \quad \sum y_i^2 = 9914.07, \quad \sum x_i y_i = 3892.8.$$

Then

$$S_{xx} = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n} = 76.1866,$$

$$S_{yy} = \sum y_i^2 - \frac{\left(\sum y_i\right)^2}{n} = 33.65, \qquad S_{xy} = -44.27.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = -0.57, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 49.8,$$

so the fitted line is $\hat{y} = 49.8 - 0.57x$.

(a) The fitted values and residuals $e_i = y_i - \hat{y}_i$ are:

Table 6.1: Fitted values and residuals.

x_i	y_i	$\hat{y}_i$	e_i
11	43.5	43.53	−0.03
15	40.0	41.25	−1.25
17	38.9	40.11	−1.21
20	38.1	38.4	−0.3
13	44.0	42.39	1.61
21	39.0	37.83	1.17

(b) The residual plot is shown below. The plot is sufficiently random, though there are too few points to draw strong conclusions.

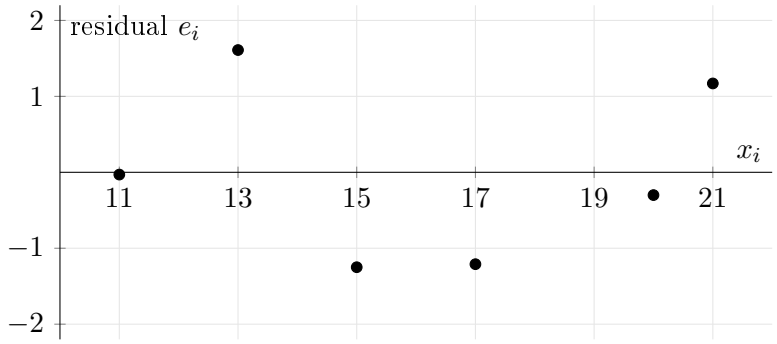


Figure 6.4: Residual plot. The scatter is sufficiently random; the sample size is small.

(c) The goodness of fit is measured by the coefficient of determination,

$$R^2 = \frac{\hat{\beta}_1 S_{xy}}{S_{yy}} = 0.75.$$

Thus about 75% of the variability in the data is explained by the model.

Chapter 7

Formula Summary

This appendix collects the principal formulas of the simple linear regression model in one place for quick reference.

7.1 Model and Assumptions

Model

$$Y = \beta_0 + \beta_1 x + \epsilon, \qquad \mathbb{E}(y \mid x) = \beta_0 + \beta_1 x, \qquad \operatorname{Var}(y \mid x) = \sigma^2.$$

The slope β_1 and intercept β_0 are the regression coefficients; ϵ is an i.i.d. error with mean zero and constant variance σ^2, assumed normally distributed.

7.2 Least Squares Estimation

Estimators

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

Corrected Sums of Squares

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum x_i\right)\left(\sum y_i\right)}{n},$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum x_i\right)^2}{n} \quad \text{(variability of } x\text{)},$$

$$S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum y_i\right)^2}{n},$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

7.3 Variance and Sums of Squares

Error Variance and Sums of Squares

$$\hat{\sigma}^2 = \frac{\mathrm{SS_E}}{n-2}, \qquad \mathrm{SS_E} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \mathrm{SS_T} - \hat{\beta}_1 S_{xy},$$

$$\mathrm{SS_T} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2, \qquad \mathrm{SS_R} = \hat{\beta}_1 S_{xy}, \qquad \mathrm{SS_T} = \mathrm{SS_R} + \mathrm{SS_E}\,.$$

7.4 Coefficient of Determination

R^2

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_E}}{\mathrm{SS_T}}.$$

Used to judge the adequacy of the model: the amount of variability in the data explained or accounted for by the model.

7.5 Correlation Coefficient

Sample Correlation Coefficient

$$R = \frac{S_{xy}}{(S_{xx}\ \mathrm{SS_T})^{1/2}}, \qquad \hat{\beta}_1 = \left(\frac{\mathrm{SS_T}}{S_{xx}}\right)^{1/2} R.$$